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The *sp* Hybridization of Tin Single Atoms for Dendrite-Free Sodium Metal Batteries

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Restricting the growth of sodium (Na) dendrites at the atomic level is the premise to enable both the stability and safety of sodium metal batteries (SMBs). Here, the universal synthesis of the fourth main group element (Sn, Ge, Pb) as single metal atoms anchored on graphene (Sn, Ge, Pb SAs/G) with *sp* hybridization for dendrite-free sodium metal anode is reported. The in situ real-time observation of Na growth on Sn SAs/G uncoils a kinetically uniform planar deposition at the atomic level for substantially suppressing the dendrite growth. The symmetrical Sn SAs/G-Na battery exhibits high Coulombic efficiency of 99.8% for 200 cycles, long-term cyclability with 600 h at 4 mA cm^{-2}, and ultralow overpotential of 40 mV at 8 mA cm^{-2}. Further, the full batteries $Na_3V_2(PO_4)_3$||Sn SAs/G-Na show unprecedented rate capability of 67 mAh g^{-1} at 20 C and a capacity retention as high as 91% after 500 cycles. The theoretical calculations of Na deposition on different M SAs/G (M=Sn, Ni, Zn, Fe, Co) reveal that the $5s^2 5p^2$ electron configuration of Sn SAs/G realizes the planar deposition mechanism of Na clusters along the graphene plane. Therefore, this work provides an atom-level accurate miniaturization strategy for constructing dendrite-free SMBs.

1. Introduction

The development of metal-based batteries to replace traditional energy sources is becoming a hot topic due to the ever-increasing energy crisis and environmental problems.[1,2] In contrast with the scarcity of lithium, sodium (Na) undoubtedly shows a unique advantage in large-scale energy storage due to its natural abundance.[3,4] Especially, sodium metal batteries (SMBs) using Na metal as anode possess the merits of high specific capacity (1166 mAh g^{-1}) and relatively low redox potential (−2.71 V vs standard hydrogen electrode).[5,6] Unfortunately, their practicability is intrinsically restricted by the formation and growth of Na dendrites during repeated charge/discharge processes, resulting in safety issues and short lifespan.[7,8] So far, various strategies such as 3D frameworks,[9,10] spatial

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confinement,[11] and solid electrolytes,[12,13] have been developed to suppress dendrite growth, however, precisely controlling the nucleation of Na metal during the initial nucleation stage is still challenging. Recently, nanoparticles with sizes of 5–20 nm of metals,[14] metal oxides,[15] sulfides,[16] and selenides[17] have been employed to guide the growth of dendrite-free Na metal. Regrettably, high-flux sodium ion flow tends to cause inhomogeneous deposition on the surface of these nanoparticle nucleating agents, leading to dendrite formation and conversion into inactive species, which restricts the further plating/stripping of Na.

To address this gap, atom-level miniaturization of such nanoparticles into single atoms is proposed to be a reliable strategy. In particular, it is reported that the atom-level transition metal sites (e.g., Zn,[18,19] Ni,[20] Co[21]) can contribute to the homogeneous nucleation and deposition of Li anodes, and maximize the electrode utilization in a weight to volume ratio. However, these transition metal single atoms have little effect on the inhibition of Na dendrites. To further enable single-atom nucleator stabilization, monolayer graphene is commonly referred to as an anchoring substrate due to its high specific surface area,[22] abundant functional groups, and defects.[23] Such features can assist the effective anchoring and 100% utilization of single atoms in the electrolyte, eventually guiding uniform metallic deposition from electrolyte. Fortunately, it is found that metallic Sn exhibits excellent affinity for Na thereby reducing the nucleation barrier,[24,25] and designing atomic Sn would offer a long-lasting and efficient Na anode.[26] Nevertheless, the fundamental mechanism investigation of the single-atom Sn to achieve dendrite-free Na metal anodes has not yet been explored.

Herein, we demonstrated for the first time that the fourth main group element of Sn, synthesized as single atoms on graphene (Sn SAs/G), could effectually suppress the growth of Na dendrite via the *sp* hybridization. The in situ growth observation revealed the kinetically uniform planar deposition of Na on Sn SAs/G, in comparison with Ni SAs/G. The Sn SAs/G anode exhibited remarkably stable plating/stripping processes of Na metal over 200 cycles and maintained an average Coulombic efficiency of 99.8%, which results in assembled full with excellent high rate capability (67 mAh g^{-1}, 20 C) and long-term cyclability (500 cycles at 1 C). Importantly, the other fourth main group elements of Ge and Pb single atoms on graphene could also inhibit Na dendrite growth, demonstrating the generality by *sp* hybridization. The first-principle calculations were further conducted to understand the influence of different single metal atoms on graphene (M SAs/G, M═Zn, Co, Ni, Fe, Sn) on Na deposition behavior, which disclosed the strong dependence of Na adsorption affinity on the nature of different single atoms. It is emphasized that the orbital structure of valence electrons in fourth main group elements guides the planar deposition of Na clusters to substantially suppress the growth of Na dendrites.

2. Results and Discussion

2.1. Synthesis and Characterizations of Sn SAs/G

To experimentally examine the electrochemical performance, we first synthesized actomially-dispersed Sn SAs/G. As shown in **Figure 1a**, graphene oxide (GO) was employed as 2D substrate to adsorb the Sn ions, where the Sn precursor was prepared by dilute hydrochloric acid (HCl) reagent to ensure its dispersibility, followed by freeze-drying the mixed solution, and subsequently an annealing process was implemented at 400 °C for 3 h in Ar gas to obtain Sn SAs/G (See details in Experimental Section). X-ray diffraction (XRD) pattern and Raman spectroscopy results confirmed the existence of graphene in Sn SAs/G (Figure S1, Supporting Information). Scanning electron microscopy (SEM) and transmission electron microscopy (TEM) images displayed the monolayer morphology of Sn SAs/G (Figure 1b,c). High-resolution TEM (HRTEM) image verified no aggregates on the surface of a sheet-like plane (Figure 1d). The atomic force microscope (AFM) image implied the flat surface of Sn SAs/G with an average ultrathin thickness of 0.57 nm (Figure 1e). The elemental mapping images clearly demonstrated the homogeneous distribution of O, C, and Sn throughout the sheet, highlighting the presence of Sn element (Figure 1f). Aberration-corrected high-angle annular dark-field STEM (HAADF-STEM) image directly disclosed the single Sn atoms (marked by red circles) with a size of ≈0.1 nm dispersed on the graphene (Figure 1g). Through inductively coupled plasma-optical emission spectrometer (ICP-OES) analysis, we inferred that the Sn content in Sn SAs/G was 2.93 wt.%. The specific surface area of Sn SAs/G was up to 543 m^2 g^{-1} (Figure S2, Supporting Information), which provided abundant nucleation sites for Na deposition. In X-ray absorption near-edge structure (XANES) spectrum, the E_0 value (the inflection point of Sn *K*-edge curve) of Sn SAs/G, SnO_2, SnO and metallic Sn foil were 29.164, 29.162, 29.145, and 29.138 keV respectively, revealing the +4 valence state of Sn in Sn SACs/G (Figure 1h), in accordance with the result of X-ray photoelectron spectroscopy (XPS, Figure S3, Supporting Information).[27,28] Fourier transformed-extended X-ray absorption fine structure (FT-EXAFS) spectrum further showed one main peak located at 1.7 Å for Sn SAs/G (Figure 1i), indicative of the single atom state of Sn. Because of the existence of O and OH groups (Figure S4, Supporting Information), FT-EXAFS simulation revealed that the Sn center atom was coordinated with four ether oxygens and one hydroxyl oxygen (inset of Figure 1j; Table S1, Supporting Information), identical to the experimental FT-EXAFS curve (Figure 1j).[29] Further, an enlarged HAADF-STEM image (Figure S5, Supporting Information) demonstrated the presence of Sn SAs located in the defective carbon sites of graphene after Na deposition and stripping process, evidencing a stable state of Sn SAs in SMBs.

2.2. In Situ Characterizations of Na Deposition

To directly observe the evolution of Na metal deposition on the Sn SAs/G, in situ electro-chemo-mechanical experiment was carried out. This measurement was operated through an environmental transmission electron microscope (ETEM) setup (**Figure 2a**). Dipped by the AFM cantilever, a Na metal covered with Na_2O layer (Na_2O/Na) acted as counter electrode while the Na_2O layer served as the solid-state electrolyte. Then, M SAs/G (M═Sn, Ni) electrode was loaded on Cu substrate (Figure S6, Supporting Information). Subsequently, the real-time observation of the Na deposition process was achieved when electron beam passed through the simulated cell. Impressively, Na metal grew uniformly on Sn SAs/G (Figure 2b,c) and there were no

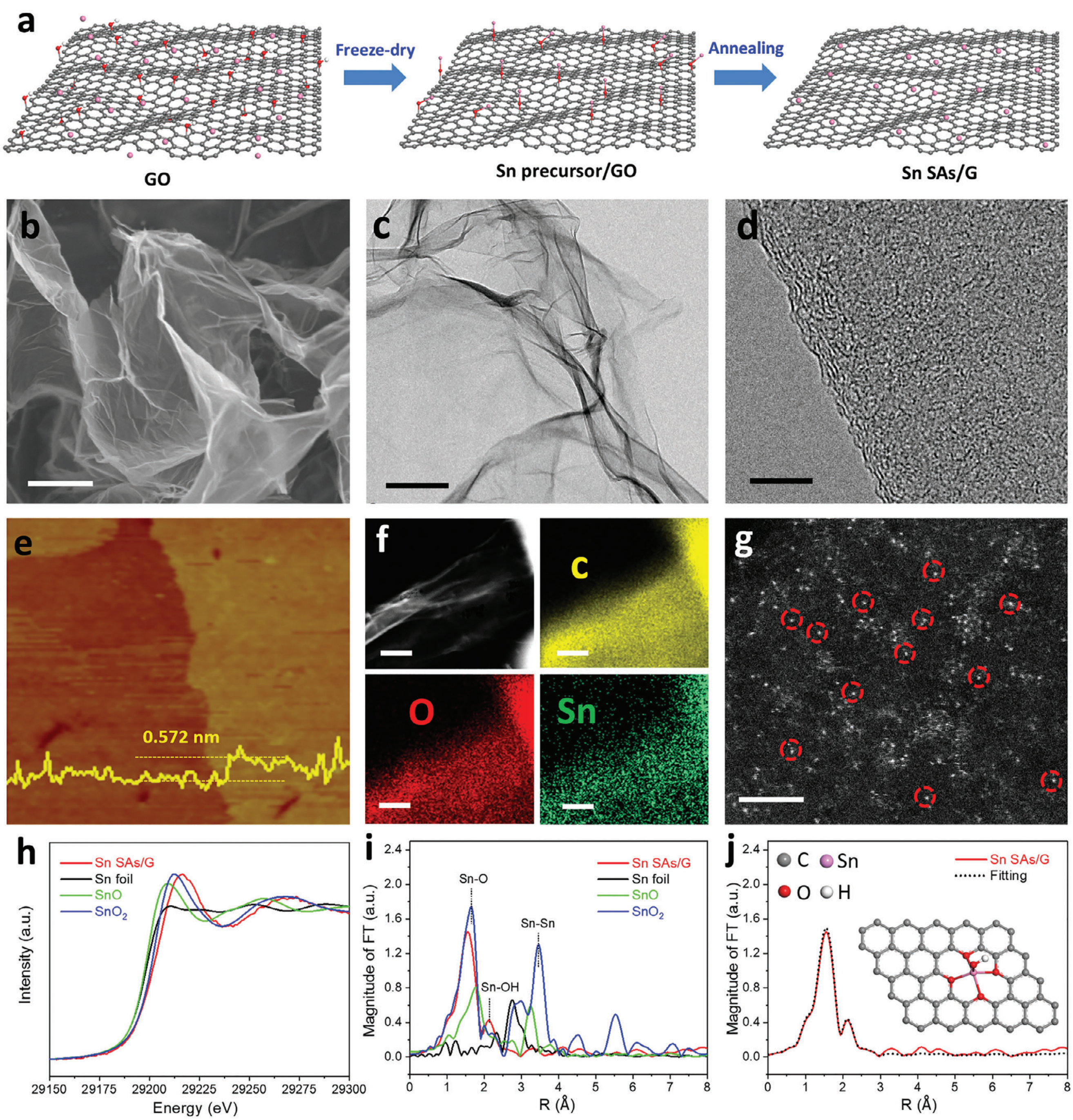

Figure 1. Synthesis and characterizations of Sn SAs/G. a) Schematic of the preparation of Sn SAs/G. b) SEM image, c) TEM image, d) HRTEM image, e) AFM image (inset in yellow line: height profile), f) STEM image and the corresponding elemental mapping (O, C, Sn), and g) HAADF-STEM image of Sn SAs/G. h) XANES and i) FT-EXAFS spectra at the Sn *K*-edge for Sn SAs/G, SnO, SnO_2, and metallic Sn foil. j) FT-EXAFS curves of the proposed Sn SAs/G structure (black line) and the measured Sn SAs/G (red line). Inset is the proposed configuration of Sn SAs/G architecture. The scale bars in (b), (c), (d), (f), and (g) are 1 µm, 100, 5, 100, and 2 nm, respectively.

precipitates of Na from Sn SAs/G even after extending the deposition time to 2797 (Figure 2d) and 4429 s (Figure 2e), further validating the uniform Na deposition on the Sn SAs/G. We also investigated the behavior of Na deposition on pure graphene (Figure 2f–i) and Ni SAs/G (Figure 2j–m). It was observed that several small pieces of Na dendrites were sprouted out at the edge of graphene after 3314 s of deposition time (red arrow, Figure 2g–i), and the Na nanoparticles in a diameter of ≈10 nm appeared on the surface of Ni SAs/G after 2252 s of deposition time (blue circle, Figure 2k–m). Consequently, this kind of epitaxial Na deposition on Sn SAs/G prevented the formation of spiny crystal nucleus and Na dendrite growth.

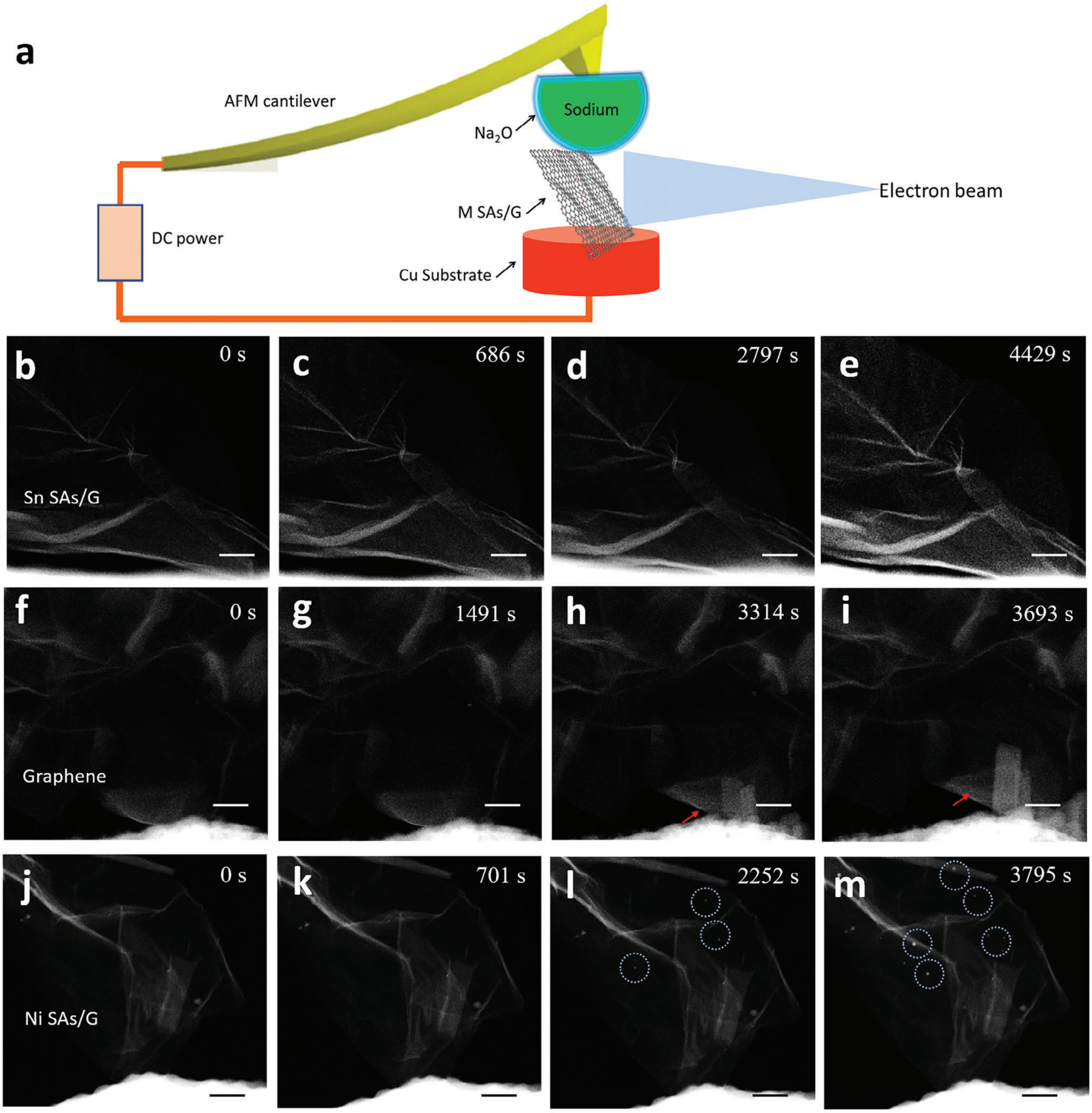

Figure 2. In situ characterizations of Na deposition kinetics on Sn SAs/G, Ni SAs/G, and graphene. a) Schematic of the AFM-ETEM set-up used for in situ observation of Na metal deposition. b–m) In situ TEM observations of the Na deposition process with (b–e) Sn SAs/G, (f–i) graphene, and (j–m) Ni SAs/G taken at different deposition times. All the scale bars in (b–m) are 200 nm.

2.3. Electrochemical Performance of Sn SAs/G Anode

To verify the reliability of the above-mentioned mechanisms, we further assembled the Sn SAs/G, Ni SAs/G, and graphene (G) anodes into half cells (Figure S7, Supporting Information), together with Na counter electrode and 1 M $NaPF_6$/diglyme electrolyte. Obviously, the single atom Sn exhibited a lower nucleation overpotential, demonstrating its sodiophilic behavior (Figure S8, Supporting Information). **Figure** 3a shows the Coulombic efficiencies of Sn SAs/G, Ni SAs/G, and G anodes with an areal capacity of 1 mAh cm^{-2} at 1 mA cm^{-2}, respectively. As demonstrated, Sn SAs/G exhibited highly stable Na plating/stripping behavior, maintaining a high average Coulombic efficiency of 99.8% for 200 cycles. From the detailed voltage curves (Figure 3b), Sn SAs/G anode maintained stable plating/stripping voltage during the 1st, 100th, and 200th cycles. In contrast, the G anode

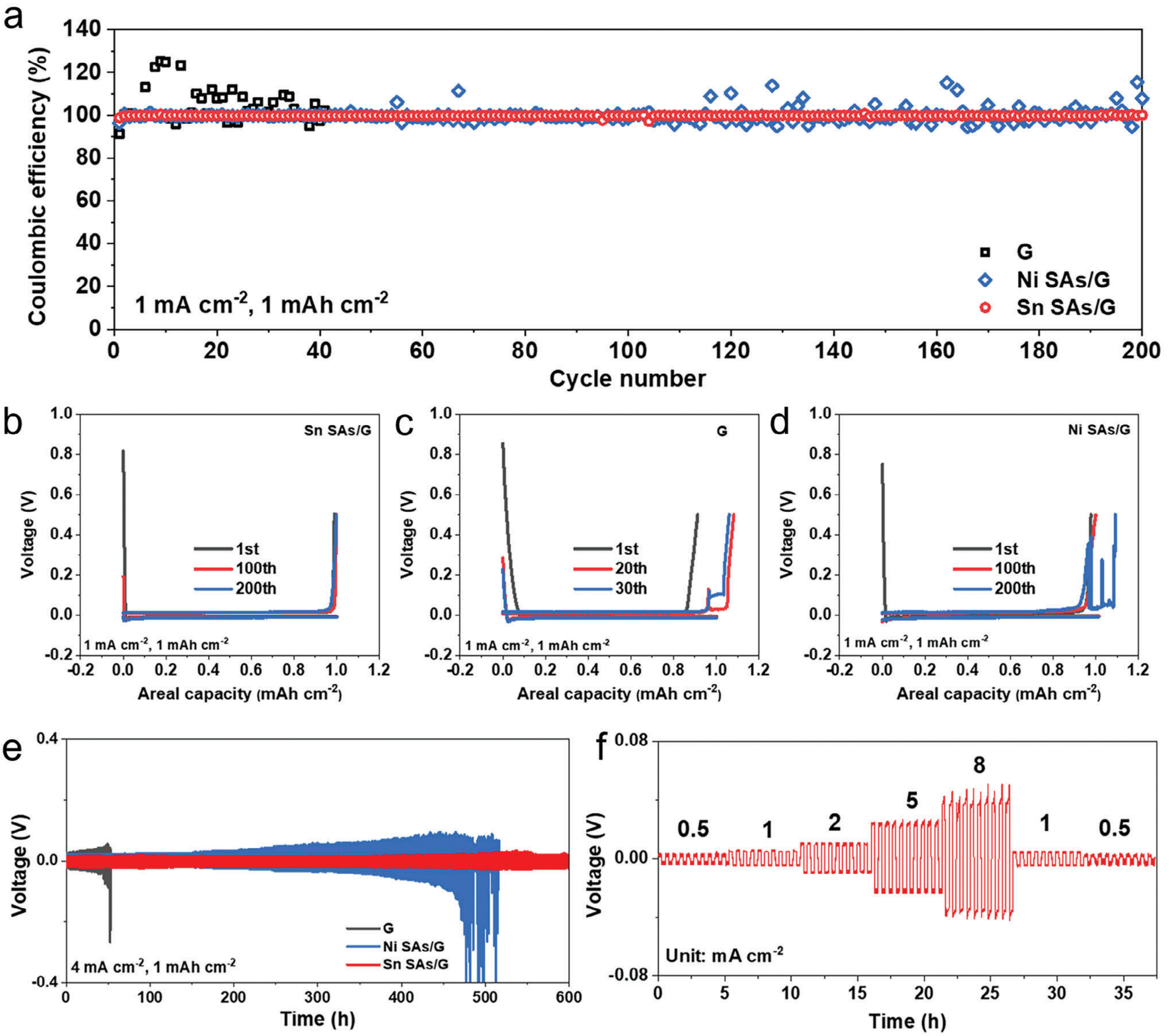

Figure 3. Electrochemical characterizations of Sn SAs/G, Ni SAs/G, and G electrodes. a) Comparison of Coulombic efficiency of Sn SAs/G, Ni SAs/G, and G electrodes with an areal capacity of 1 mAh cm^{-2} at 1 mA cm^{-2}. b–d) Charge-discharge voltage profiles of different cycles of Na plating/stripping on b) Sn SAs/G, c) G, and d) Ni SAs/G electrodes at 1 mA cm^{-2} for 1 h. e) Voltage profiles of Sn SAs/G, Ni SAs/G, and G electrodes charging/discharging at 4 mA cm^{-2} for 15 min. f) Voltage profiles of Sn SAs/G electrodes plating/stripping at varying current densities of 0.5, 1, 2, 5, 8, 1, and 0.5 mA cm^{-2} for 15 min.

showed an unstable Coulombic efficiency during the 40 cycles (Figure 3a) and fluctuant voltage curves (Figure 3c), suggestive of the formation of fragile solid electrolyte interface (SEI) layer and Na dendrites on the G anode.[30] Compared with the G anode, Ni SAs/G anode showed slightly improved Coulombic efficiency during the first 100 cycles. However, from the 100th to 200th cycles, the Coulombic efficiency and voltage profile fluctuated greatly (Figure 3d).

Furthermore, to evaluate the effect of Sn SAs on preventing dendrite formation, Sn SAs/G-Na||Na batteries were used for Na plating/striping, through electrodeposition at 0.5 mA cm^{-2} for 16 h to obtain a Sn SAs/G-Na electrode. Figure 3e shows the voltage curves of symmetrical Sn SAs/G-Na battery with a capacity of 1 mAh cm^{-2} at 4 mA cm^{-2}. Notably, the enlarged voltage profiles at 50 ~54, 300 ~ 304, 500 ~ 504, and 596 ~ 600 h demonstrated an ultra-low initial overpotential of only 17 mV (Figure S9, Supporting Information). Besides, no large fluctuation appeared within the recorded 600 h, validating stable cycling performance. In comparison, both symmetrical G-Na (45 mV) and Ni SAs/G-Na (35 mV) batteries showed not only higher initial overpotentials but also gradually increased overpotential after the continuous Na plating/striping process. Moreover, there were severe voltage fluctuations after a certain cycling time (50 h for G-Na and 450 h for Ni SAs/G-Na), due to repeated SEI

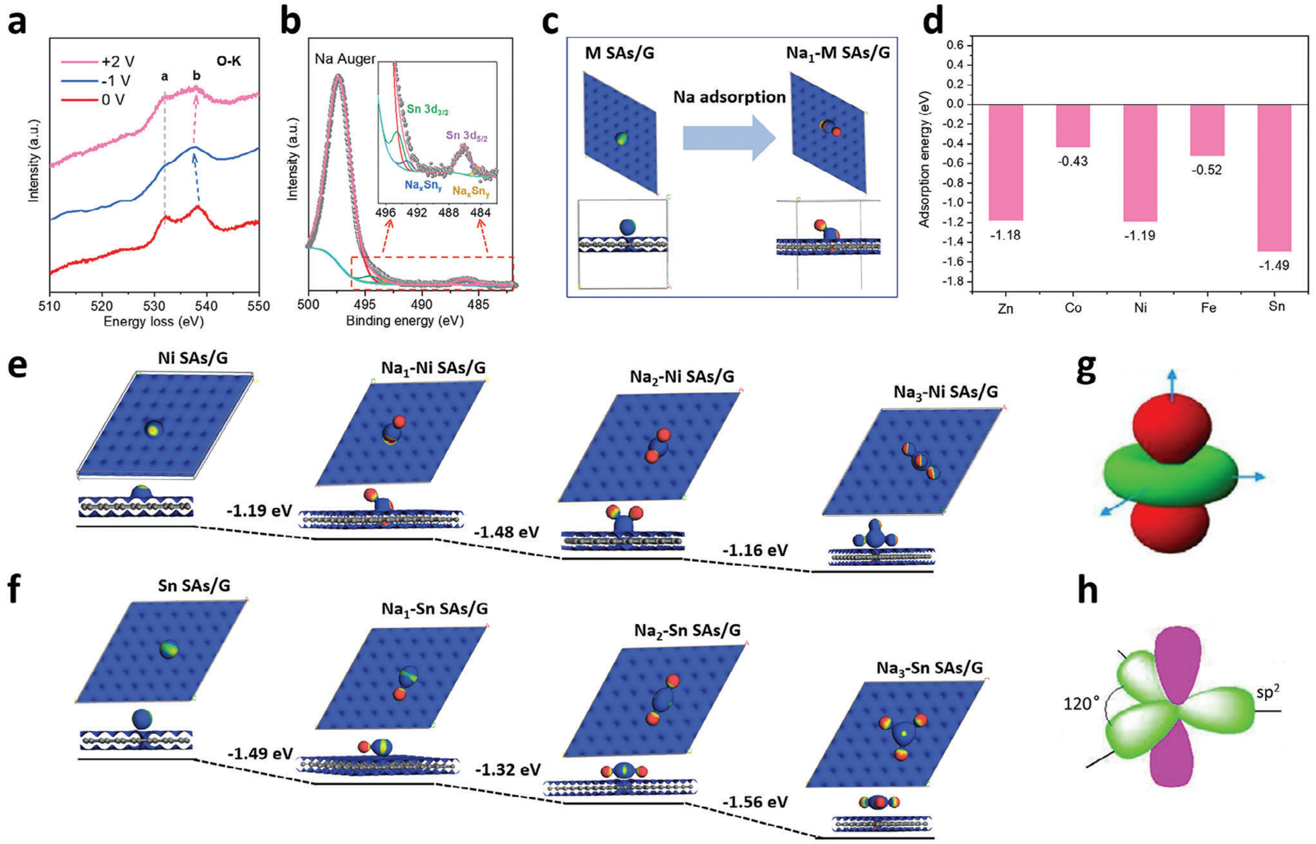

Figure 4. Ex situ characterizations and theoretical calculation of the chemisorption behaviors of Na ions towards M SAs/G (M = Zn, Co, Ni, Fe, Sn). a) In situ EELS spectra of the O-*K* edge for the Sn SAs/G electrode after Na plating/stripping on various voltages. b) The Sn 3d XPS spectrum for the Sn SAs/G electrode after alloying reaction. c) The configuration evolution of single atoms anchored on graphene (M SAs/G) after Na adsorption. d) The Na adsorption energy of different single atoms (Zn, Co, Ni, Fe, Sn) anchored on the C_2 sites of graphene. The energy profiles, and configurations of e) Ni SAs/G and f) Sn SAs/G adsorbed with one, two, and three Na ions. The valence electrons hybrid configurations of (g) Ni SAs and (h) Sn SAs.

fracture and formation.[31,32] Notably, symmetrical Sn SAs/G-Na batteries displayed excellent rate capability (Figure 3f). The overpotentials of Na stripping/plating were ≈3, 5, 10, 23, and 40 mV at 0.5, 1, 2, 5, and 8 mA cm^{-2}, respectively. For Sn SAs/G-Na anode, the maximum withstand current density of 8 mA cm^{-2} was much higher than those of recently reported works recorded in Figure S10 and Table S2 (Supporting Information), such as organic electrolyte stabilized Na metal anodes (0.1 mA cm^{-2} of $NaCF_3SO_3$ in triethylene glycol dimethyl ether electrolyte,[33] 2 mA cm^{-2} of $NaPF_6$ in dimethoxyethane[34]), surface modified Na metals (1 mA cm^{-2} of NaBr coated Na metal,[35] 5 mA cm^{-2} of Al_2O_3 coated Na metal[36,37]), and 3D conductive network composite/Na metal anodes (5 mA cm^{-2} of Fe_2O_3 coated carbon textile,[38] 0.5 mA cm^{-2} of carbon microspheres[39]). The overpotential returned to 5 and 3 mV again when the current density was reduced to 1 and 0.5 mA cm^{-2}, respectively, validating excellent reversibility. To highlight the critical role of Sn SAs, we also designed the anode of SnO_2 nanoparticles loaded on graphene (SnO_2/G, Figure S11, Supporting Information) for comparison. As shown in Figure S12 (Supporting Information), the Coulombic efficiency of SnO_2/G electrode was extremely low and dropped to 0 after ≈50 cycles, which possibly was caused by a short-circuit of the cell. Furthermore, the overpotential of symmetrical SnO_2/G-Na||Na battery fluctuated drastically after only cycling for 25 h and a short circuit occurred at 30 h. The far interior performance of SnO_2/G demonstrated that the Sn SAs/G could effectively inhibit the growth of Na dendrites, and boost the stability of Na metal anode.

2.4. Mechanism Revealing of M SAs/G for Na Deposition

The cyclic voltammogram (CV) curves showed that there was a pair of redox peaks in the Sn SAs/G (Figure S13a, Supporting Information). Meanwhile, the half cell assembled by SnO_2/G exhibited an additional reduction peak of ≈0.25 V (Figure S13b, Supporting Information), corresponding to the formation of the alloy.[26,40] Moreover, in situ electron energy loss spectroscopy (EELS) can significantly substantiate that the two characteristic peaks (a and b in **Figure 4a**) of the O *K* edge in Sn SAs/G displayed a narrower energy interval and the intensity weakened during discharging.[41,42] Subsequently, the peak position and intensity recovered after the charging, suggesting that the valence state of Sn first decreased and then increased during the Na plating/stripping processes. The Sn 3d XPS spectrum of Sn SAs/G electrode indicates the existence of Na-Sn alloy after Na deposition. In addition to the peak at ≈497.2 eV belonging to Na KLL, the peaks locked at ≈494.4 and ≈486.2 eV marked the presence of metallic Sn (Figure 4b). Importantly, the two splitting peaks of Sn $3d_{3/2}$ and Sn $3d_{5/2}$ appearing at ≈492.5 and ≈ 484.1 eV correspond to the Na_xSn_y alloy.[43,44] Additionally, the results of the SnO_2 electrode revealed that the peak intensity of Sn and Na-Sn alloy became more pronounced as the Sn content increased (Figure S14, Supporting Information).

To illustrate the kinetics of Sn-based Na metal batteries and the differences of various single metal atoms in Na deposition, density functional theory (DFT) calculations were conducted to simulate the chemisorption of Na^+ towards M SAs/G (M=Zn, Co, Ni, Fe, Sn). First, various adsorption positions were checked to identify the most stable position of single atoms on the monolayer graphene. Here, graphene was chosen as the scaffold to stabilize the single atoms,[45] which contained carbon sites bond to three carbon atoms (marked as C_1), carbon sites bond to two carbon atoms (marked as C_2) and they were the main reaction sites of graphene under the working condition (Figure S15, Supporting Information). Taking Sn single atoms (Sn SAs) as an example, it was disclosed that the adsorption energy of Sn SAs decorated on C_1 and C_2 was 0.093 and −4.907 eV, respectively (Figure S16, Supporting Information), manifesting that C_2 sites were energetically preferred for anchoring single atoms. Then, Na atom was bonded to various metal single atoms anchored on the C_2 sites of graphene (M SAs/G, M=Sn, Ni, Zn, Fe, Co) and the corresponding sequence of Na adsorption energies were −1.49 eV for Sn SAs/G, −1.19 eV for Ni SAs/G, −1.18 eV for Zn SAs/G, −0.52 eV for Fe SAs/G, and −0.43 eV for Co SAs/G, respectively (Figure 4c,d). To further distinguish the behavior of Na crystal nucleus growth, Ni SAs/G and Sn SAs/G with relatively large Na adsorption energies were chosen for simulation (Figure 4d), in which the most stable adsorption configurations and corresponding adsorption energies of Ni SAs/G and Sn SAs/G bond with one, two and three Na atoms were presented (Figure 4e,f). For one, two and three Na atoms anchored on single metal atoms, the energies were −1.19 eV for Na_1-Ni SAs/G, −2.67 eV for Na_2-Ni SAs/G, and −3.83 eV for Na_3-Ni SAs/G while −1.49 eV for Na_1-Sn SAs/G, −2.81 eV for Na_2-Sn SAs/G, and −4.37 eV for Na_3-Sn SAs/G, respectively. Apparently, Sn SAs/G delivered better Na affinity not only for one Na atom but also for Na nanoclusters, compared with the corresponding adsorption configurations of Ni SAs/G. The most stable Na_3 nanocluster adsorbed on Ni SAs/G was in the form of the dendrite and spherical morphologies (Figure S17, Supporting Information), which was caused by the isotropic $3d^8 4s^2$ electron configuration of Ni (Figure 4g), resulting in the randomness growth of Na atoms. As a result, this kind of protruding Na_3 nucleus tended to induce the tip discharge, and introduced the local current concentration, thereby yielding the Na dendrite and subsequent short circuit.[20] In a sharp contrast, the Na nanocluster in Na_3-Sn SAs/G was in the form of 2D morphology. It should be emphasized that 2D Na_3 nanocluster was the only thermodynamically stable structure in the presence of Sn SAs/G (Figure S18, Supporting Information). As a fourth main group element, the electrons in the $5s^2 5p^2$ orbit of Sn atoms would form a sp^2 hybridization (Figure 4h), resulting in the anisotropy of Na atoms adsorption. Subsequently, the Na deposition will preferentially grow around $SnNa_3$ clusters, while 2D structured $SnNa_3$ clusters induce Na to preferentially deposit along the plane of graphene, rather than growing perpendicular to graphene to form the dendrites. Consequently, the unique valence electron distribution of the fourth main group elements restrained the formation of protruded Na at atomic level.

2.5. Performance of Sn SAs/G Based Full Battery

To demonstrate the applicability of Sn SAs/G-Na anode, a full battery (($Na_3V_2(PO_4)_3$ (NVP)||Sn SAs/G-Na) was assembled based on Sn SAs/G-Na (Na deposition with 3 mAh cm^{-2}, Figures S19 and S20, Supporting Information) as anode, and NVP with the carbon coating as cathode, as schematically shown in **Figure 5a**. HRTEM image (Figure S21, Supporting Information) displayed the *d*-spacing of 0.37 nm corresponding to the (113) plane of NVP, and ≈3 nm carbon layer coated on the surface of NVP. By analyzing CV curves obtained from 0.1 to 1.5 mV s^{-1}, the fast reaction kinetics process of the NVP||Sn SAs/G-Na batteries was revealed (Figure S22, Supporting Information), showing extremely small potential polarization of 0.2 V (at 0.1 mV s^{-1}), and large Na ion diffusion coefficient (D_{Na+}) of 1.55×10^{-10} cm^2 s^{-1} for oxidation peak and 1.61×10^{-10} cm^2 s^{-1} for reduction peak. Furthermore, galvanostatic charge and discharge profiles of NVP||Sn SAs/G-Na battery exhibited stable voltage plateau and low polarization voltages of ≈0.07, 0.10, 0.14, 0.24, 0.27, and 0.33 V at increasing current rates of 1, 2, 5, 10, 15, and 20 C, respectively (Figure 5b), manifesting superior conductivity and structural stability of Sn SAs/G anode. It is noted that the two discharge platforms of NVP||Sn SAs/G-Na battery were attributed to a local redox environment rearrangement from the ubiquitous transfer of Na (1) to Na (2) existed in NASICON-type materials.[46] More importantly, NVP||Sn SAs/G-Na full battery displayed outstanding rate capability with high capacity of 98, 97, 91, 83, 78, and 67 mAh g^{-1} at various current rates of 1, 2, 5, 10, 15, and 20 C, respectively (Figure 5b,c). A large capacity could be recoverable to 98 mAh g^{-1} when the current density was switched back to 1 C.

Further, the cycling stability of NVP||Sn SAs/G-Na battery was compared with NVP||Ni SAs/G-Na, NVP||G-Na and NVP||bare Na (NVP||Na) metal batteries (Figures 5d and S23 and S24, Supporting Information). It is revealed that the capacity of NVP||G-Na batteries reduced rapidly as the cycle progressed (Figure S23, Supporting Information), while NVP||Ni SAs/G-Na battery was relatively stable during the initial stage, but rapidly attenuated after 150 cycles, ascribing to the growth of Na dendrite. In sharp contrast, NVP||Sn SAs/G-Na battery delivered a high discharge capacity of 102 mAh g^{-1} for the first cycle, and still maintained 93 mAh g^{-1} after 500 cycles, showing an extremely low average capacity decay rate per cycle of only 0.018%, demonstrative of extraordinary long-term stability. More notably, when the other fourth main group elements of Ge and Pb synthesized as single atoms on graphene (Ge SAs/G, Pb SAs/G, Figures S25 and S26, Supporting Information), both Ge SAs/G and Pb SAs/G, similar to Sn SAs/G, could exceptionally suppress the formation of Na dendrite (Figure S27, Supporting Information), demonstrating wide universality of fourth main group elements for Na dendrite inhibition.

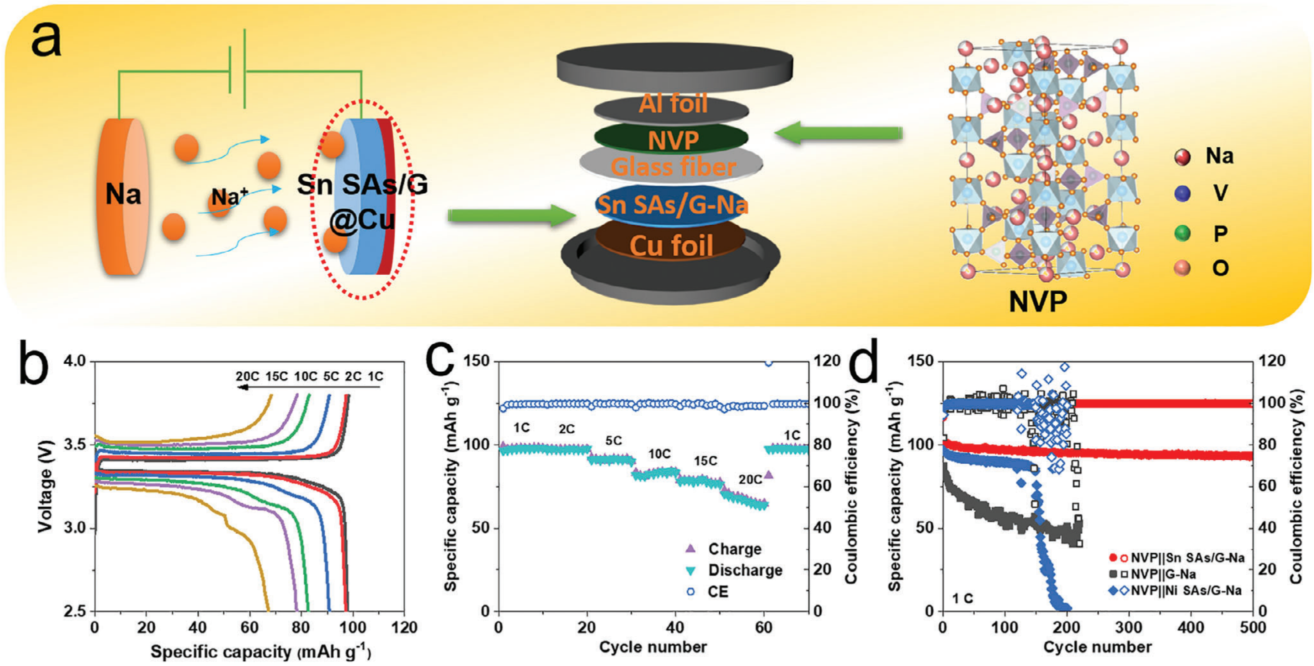

Figure 5. Electrochemical characterization of NVP||Sn SAs/G-Na full batteries. a) Schematic of NVP||Sn SAs/G-Na battery. b) Galvanostatic charge and discharge profiles and c) rate capability of NVP||Sn SAs/G-Na battery obtained at different rates. d) Cycling stability of NVP||Sn SAs/G-Na, NVP||G-Na and NVP||Ni SAs/G-Na measured at 1 C.

3. Conclusion

In summary, we demonstrated the fourth main group elements with *sp* hybrization as single stoms on graphene for dendrite free SMBs, and both the in situ TEM observation and theoretical simulation disclosed the fundamental mechanism of Na metal uniform planar deposition in atomic level on Sn SAs/G without interfacial dendrite formation and Na aggregation. Furthermore, the symmetrical Sn SAs/G-Na battery exhibited high Coulombic efficiency of 99.8% for 200 cycles, long-term stability with 600 h at 4 mA cm^{-2} and ultra-low overpotential of 40 mV at 8 mA cm^{-2}. Our NVP||Sn SAs/G-Na batteries showed high capacity of 67 mAh g^{-1} at 20 C and long-term cyclability. Similarly, the Ge SAs/G and Pb SAs/G could also inhibit the dendrite growth of Na, manifesting wide universality. It is theoretically calculated that Sn single atoms possessed strong Na adsorption energy of −1.49 eV, and unique *sp* orbital hybridization of the fourth main group elements guided the planar growth of Na_3 nanoclusters. Therefore, this work sheds new insights on the development of the fourth main group elements in single atom form for high-performance dendrite free SMBs.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Keywords

graphene, planar deposition mechanism, Sn single atoms, sodium metal batteries, sp hybridization

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